

Practical 11 - Estimating causal effects using observational data 2

Survival Analysis

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Datasets required: `rotterdam_p11`, `rotterdam_p10`

R packages required: `survival`, `pec`, `adjustedCurves`, `riskRegression`

```
library(survival)
library(pec)
library(adjustedCurves)
library(riskRegression)
```

Introduction

In this practical we will continue to use the `rotterdam` data used in the lectures and in Practical 10. In Part A we will use a modified version of the data (`rotterdam_p11.csv`) in which some individuals change their hormone therapy use over time. In Part B we will return to using the data from Practical 10 (`rotterdam_p10.csv`), in which treatment status is fixed over time, and in which we will focus on competing risks outcomes.

Aims

The aims of this session are:

- Estimate conditional and marginal survival probabilities at a specific time under different treatment options, when observed treatment status changes over time.
- Estimate causal effects of a treatment in terms of marginal risk differences, risk ratios and restricted mean survival times, when observed treatment status changes over time.
- Estimate marginal survival curves under different treatment options, when observed treatment status changes over time.
- Estimate causal effects of treatments on competing events.

Part A: Changes in treatment status over time

We will use a modified version of the data, called `rotterdam_p11`. In this data some individuals not on hormone therapy at time 0 switch onto it at a later time. As in the previous practical the focus is on the composite event of disease recurrence or death from any cause, and on follow-up to 5 years. Administrative censoring at 5 years has already been applied. Variables used in this practical are summarised in the table below. We will also use the same set of adjustment variables as used in Practical 10 (`age`, `meno`, `size`, `grade`, `enodes`, `pgr`, `er`, `year_cat`, `chemo`).

This is not an official version of the data - we created it to illustrate the methods.

Variable name	Description
<code>pid</code>	Patient identifier
<code>hormon</code>	Ever used hormonal treatment: 0=no, 1=yes
<code>t_hormon</code>	Time dependent hormone treatment use: 0=no, 1=yes
<code>outcome</code>	Indicator of composite outcome: 0=censored, 1=event
<code>start_time</code>	Start time for this time interval
<code>stop_time</code>	Stop time for this time interval. Time of censoring or the composite event.
<code>rownum</code>	Row number for a given individual (1 or 2)
<code>nrow</code>	Ttal number of rows for a given individual (1 or 2)

Load the data:

```
rott2<-read.table("datasets/rotterdam_p11.csv",sep=";",header=T)
```

Question 1

- Take a look at the format of the data. How many rows of data are there and how many individuals?
- How many people switch onto hormone therapy over time (i.e. from `hormon=0` in their row 1 to `hormon=1` in their row 2)?

Question 2

In this question we will use the artificial censoring approach to handle the treatment switching and to estimate conditional survival probabilities under the treatment strategies of: (i) Start hormone therapy directly following surgery (and continue), (ii) Do not start hormone therapy directly following surgery or for the duration of follow-up.

Use the code below to implement the following:

- Modify the data so that people who switch from `hormon=0` to `hormon=1` are censored at the time of treatment switching. How many rows of data are there now?
- Fit a Cox model with hormone therapy as your main exposure, and conditioning on all the same set of variables as in the previous practical (`age`, `meno`, `size`, `grade`, `enodes`, `pgr`, `er`, `year_cat`, `chemo`).
- Interpret the hazard ratio estimate for the hormone therapy variable.

```
#data set with censoring at initiation of hormone therapy
#for people who switch from hormon=0 to hormon=1
roott2.cens<-roott2[(roott2$hormon==1 & roott2$rownum==1) |
                    (roott2$hormon==0),]

#Fit Cox model to the modified data
cox.adj<-coxph(Surv(stop_time,outcome)~hormon+age+meno+size+
               as.factor(grade)+enodes+pgr+er+year_cat+chemo,
               data=roott2.cens,x=TRUE)
```

Question 3

- What has been assumed about censoring in the Cox model fitted in Question 2?
- Under what assumptions can the HR for hormone therapy be interpreted as the causal effect of treatment strategy (i) versus (ii) on the outcome.

Question 4

- Using the adjusted Cox model from question 2, obtain an estimate of the risk of recurrence or death up to 5 years for each individual in the data under the two strategies of interest: (i) Start hormone therapy directly following surgery (and continue), (ii) Do not start hormone therapy directly following surgery or for the duration of follow-up.
- For person 1, what would the estimated risk of recurrence or death up to 5 years have been if they had followed treatment strategies (i) and (ii)?

Question 5

- Using your model from Question 4, obtain estimates of the marginal risks up to time 5, $\Pr(T \leq 5 | do(X = 1))$ and $\Pr(T \leq 5 | do(X = 0))$, where $X = 1$ denotes treatment strategy (i) and $X = 0$ denotes treatment strategy (ii).
- Obtain estimates of the marginal risk difference and marginal risk ratio and interpret your results.

You can do this ‘by hand’ using the conditional survival probabilities obtained in Question 4, or using `adjustedsurv`.

Part B: Competing events

In this part return to using the data from Practical 10 (`rotterdam_p10.csv`), in which treatment status is fixed over time. We now consider the outcomes of ‘disease recurrence’ and ‘death without recurrence’ as competing events. The variables were summarised in the table at the start of Practical 10.

Question 1

- Load the data. The variable `time` is years to the first of recurrence or death without recurrence. The variable `outcome_cr` indicates what happened to the person at the end of their follow-up: 0=censored, 1=recurrence, 2=death without recurrence
- How many people had each of the two competing events?
- Obtain a plot of the Aalen-Johansen estimates of the cumulative incidences of the two event types by hormone therapy use. You can look back to Practical 6 (Question 6) for code to do this.

Question 2

In this question we will estimate conditional cumulative incidences of the two event types under hormone therapy use and non-use for each individual in the data. Use the code below to do the following:

- Fit cause-specific Cox models for each of the two event types of interest, including hormone therapy as the main exposure and adjusting for the same set of variables used above.
- Use `survfit` to obtain estimates of the cumulative incidences for the two event types for each individual under the two treatment strategies (setting `hormon=1` and `hormon=0`).
- Take the means across individuals to obtain the marginal cumulative incidences under the two treatment strategies.

Interpret the resulting plot.

```
#---  
#Fit cause-specific Cox models  
cox.adj.cr<-coxph(Surv(time,as.factor(outcome_cr))~hormon+age+meno+  
                  size+as.factor(grade)+enodes+pgr+er+year_cat+chemo,
```

```

data=roth,id=pid)

#---
#Conditional cumulative incidences under each treatment strategy

#create dataset in which hormon is set to 1 for everyone
data.hormon1<-roth
data.hormon1$hormon<-1

#create dataset in which hormon is set to 0 for everyone
data.hormon0<-roth
data.hormon0$hormon<-0

#obtain cumulative incidences for each individual given their covariates
#under each treatment strategy
coxpred.1 <- survfit(cox.adj.cr, newdata= data.hormon1)
coxpred.0 <- survfit(cox.adj.cr, newdata= data.hormon0)

#cumulative incidence for recurrence (pstate[,2])
#under each treatment strategy
#at all event times
cinc.recur.1.individual<-coxpred.1$pstate[,2]
cinc.recur.0.individual<-coxpred.0$pstate[,2]

#cumulative incidence for death without recurrence (pstate[,3])
#under each treatment strategy
#at all event times
cinc.death.1.individual<-coxpred.1$pstate[,3]
cinc.death.0.individual<-coxpred.0$pstate[,3]

#---
#Marginal cumulative incidences under each treatment strategy
#at each event time

cinc.recur.1<-rowMeans(cinc.recur.1.individual)
cinc.recur.0<-rowMeans(cinc.recur.0.individual)

cinc.death.1<-rowMeans(cinc.death.1.individual)
cinc.death.0<-rowMeans(cinc.death.0.individual)

#---
#plot marginal cumulative incidence curves

```

```

times<-coxpred.1$time#list of all event times

plot(times,cinc.recur.0,type="s",col="black",lty=1,lwd=2,
      xlim=c(0,5),ylim=c(0,1),
      xlab="Time (years)", ylab="Cumulative incidence")
lines(times,cinc.recur.1,type="s",col="grey",lty=1,lwd=2)
lines(times,cinc.death.0,type="s",col="black",lty=2,lwd=2)
lines(times,cinc.death.1,type="s",col="grey",lty=2,lwd=2)
legend("topleft", c("Recurrence: No hormone therapy",
                    "Recurrence: Hormone therapy",
                    "Death without recurrence: No hormone therapy",
                    "Death without recurrence: Hormone therapy"),
      col=c("black","grey","black","grey"),
      lty=c(1,1,2,2), lwd=2, bty='n')

```

Question 3

The marginal cumulative incidences can also be obtained in the `adjustedCurves` package using `adjustedcif`. Take a look at the code below for doing this.

Note this this requires the cause-specific Cox models to be fitted using CSC in the `riskRegression` package.

```

#---
#Fit cause-specific hazard models using CSC from the package riskRegression
#`hormon` needs to be a factor variable
roott$hormonfac<-as.factor(roott$hormon)
cox.adj.cr2<-CSC(
  Hist(time, outcome_cr)~hormonfac+age+meno+size+as.factor(grade)+
  enodes+pgr+er+year_cat+chemo,data=roott)

#---
#Now use adjustedcif to get the standardised cumulative incidence curves
#This has to be done separately for each event type

#recurrence
adjcif1 <- adjustedcif(data=roott,
                      variable="hormonfac",
                      ev_time="time",
                      event="outcome_cr",
                      method="direct",

```

```

        outcome_model=cox.adj.cr2,
        cause=1,
        conf_int=TRUE)
plot(adjcif1, conf_int=TRUE,custom_colors=c("black","grey"),
     xlab="Time (years)",legend.title="Hormone therapy",
     ylim=c(0,0.5),title = "Disease recurrence")

#Death without recurrence
adjcif2 <- adjustedcif(data=roth,
                      variable="hormonfac",
                      ev_time="time",
                      event="outcome_cr",
                      method="direct",
                      outcome_model=cox.adj.cr2,
                      cause=2,
                      conf_int=TRUE)
plot(adjcif2, conf_int=TRUE,custom_colors=c("black","grey"),
     xlab="Time (years)",legend.title="Hormone therapy",
     ylim=c(0,0.05),title = "Death without recurrence")

```

We can also use `adjustedcif` to obtain cumulative incidence estimates at a specific time, as well as risk differences and risk ratios. This is shown below for time 5.

```

#Recurrence
adjcif1.time5 <- adjustedcif(data=roth,
                           variable="hormonfac",
                           ev_time="time",
                           event="outcome_cr",
                           method="direct",
                           outcome_model=cox.adj.cr2,
                           cause=1,
                           conf_int=TRUE,
                           times = 5)

adjcif1.time5$ate_object$meanRisk
adjcif1.time5$ate_object$diffRisk
adjcif1.time5$ate_object$ratioRisk

```

```

#Death without recurrence
adjcif2.time5 <- adjustedcif(data=roth,
                           variable="hormonfac",
                           ev_time="time",
                           event="outcome_cr",

```

```
method="direct",  
outcome_model=cox.adj.cr2,  
cause=2,  
conf_int=TRUE,  
times = 5)  
adjcif2.time5$ate_object$meanRisk  
adjcif2.time5$ate_object$diffRisk  
adjcif2.time5$ate_object$ratioRisk
```